

Status of the polarizer ring

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Overview

Current Baseline

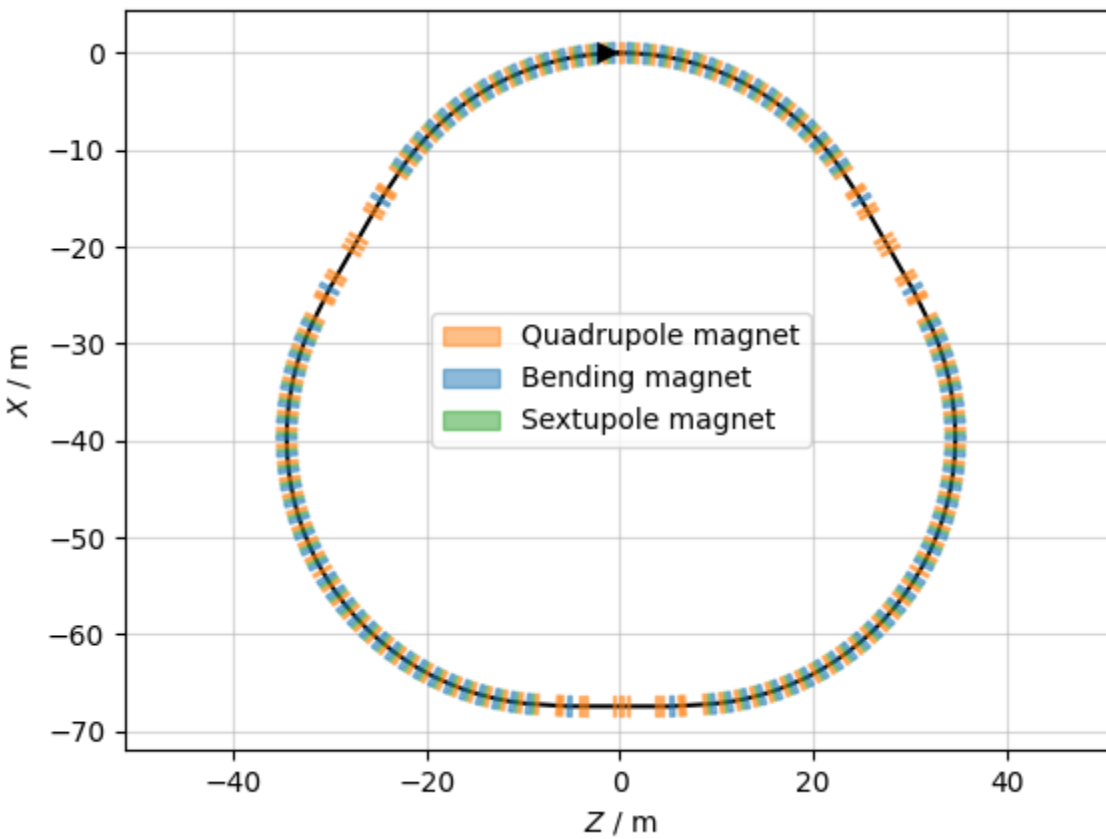
- Ring optics
- Working point
- Misalignments and orbit correction
- Dynamic aperture and momentum acceptance
- Spin tracking and equilibrium polarization
- Injection efficiency

Design Variants

- 120° phase advance
- Racetrack design

Conclusions and next steps

Ring optics

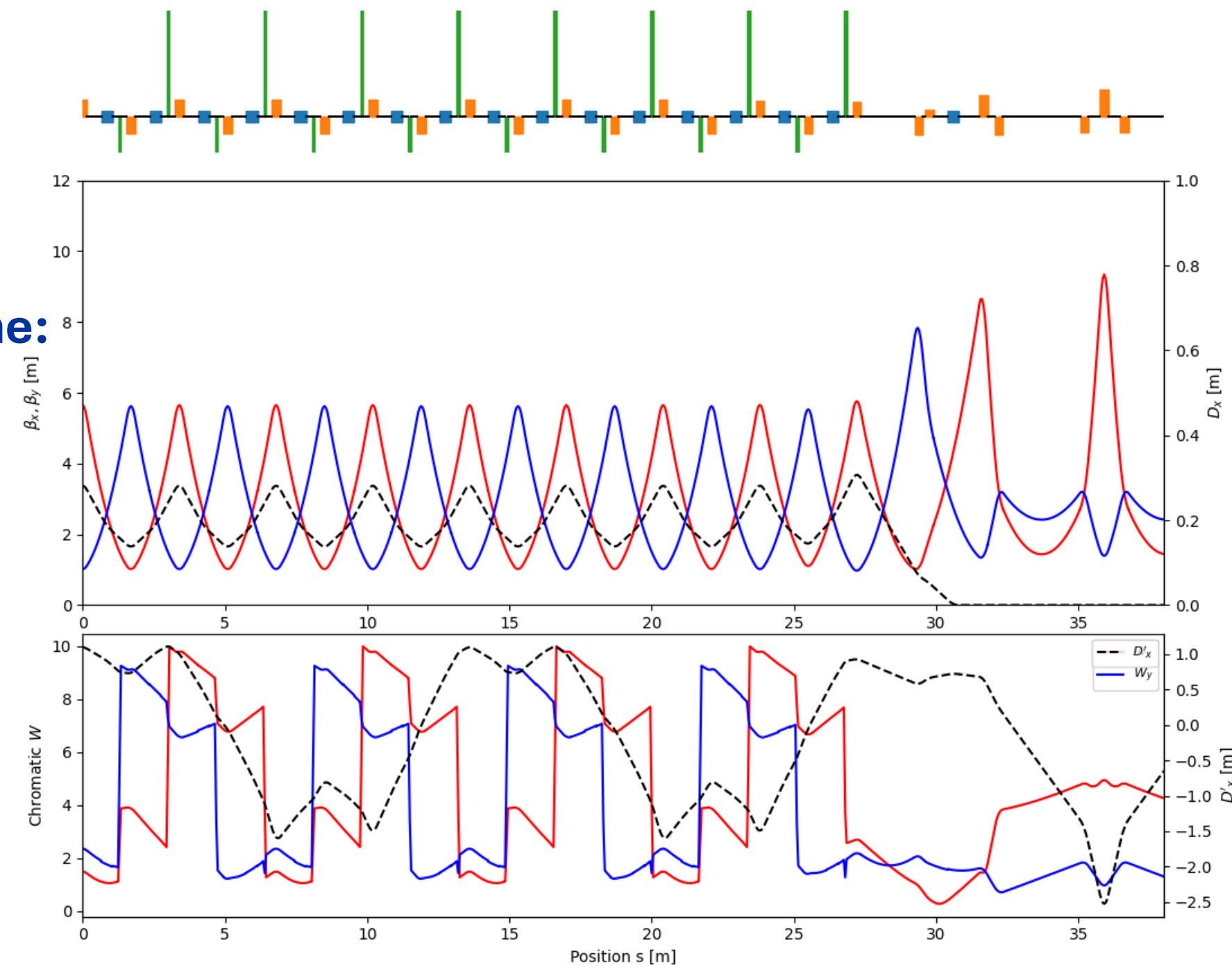


Ring parameters

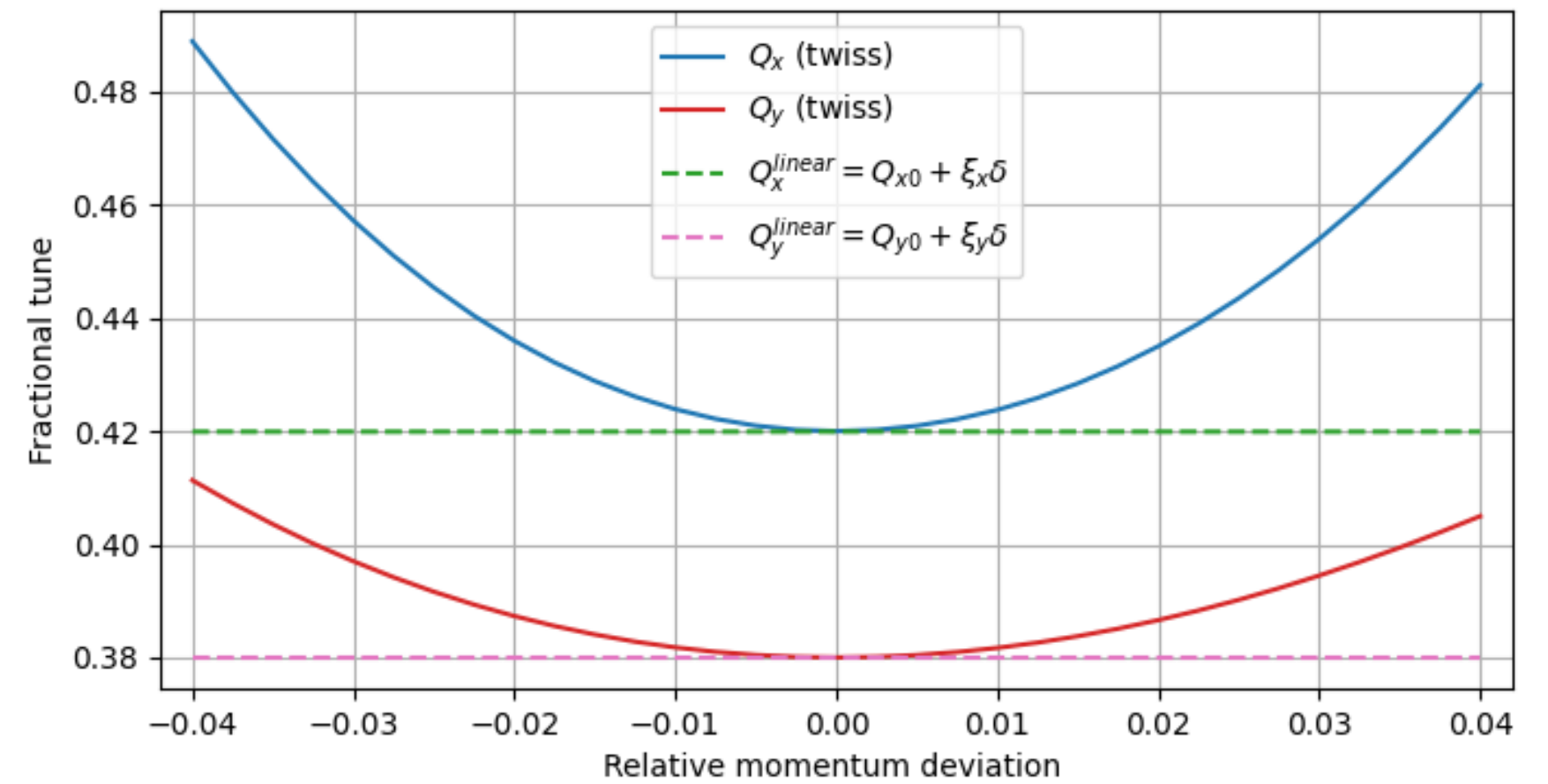
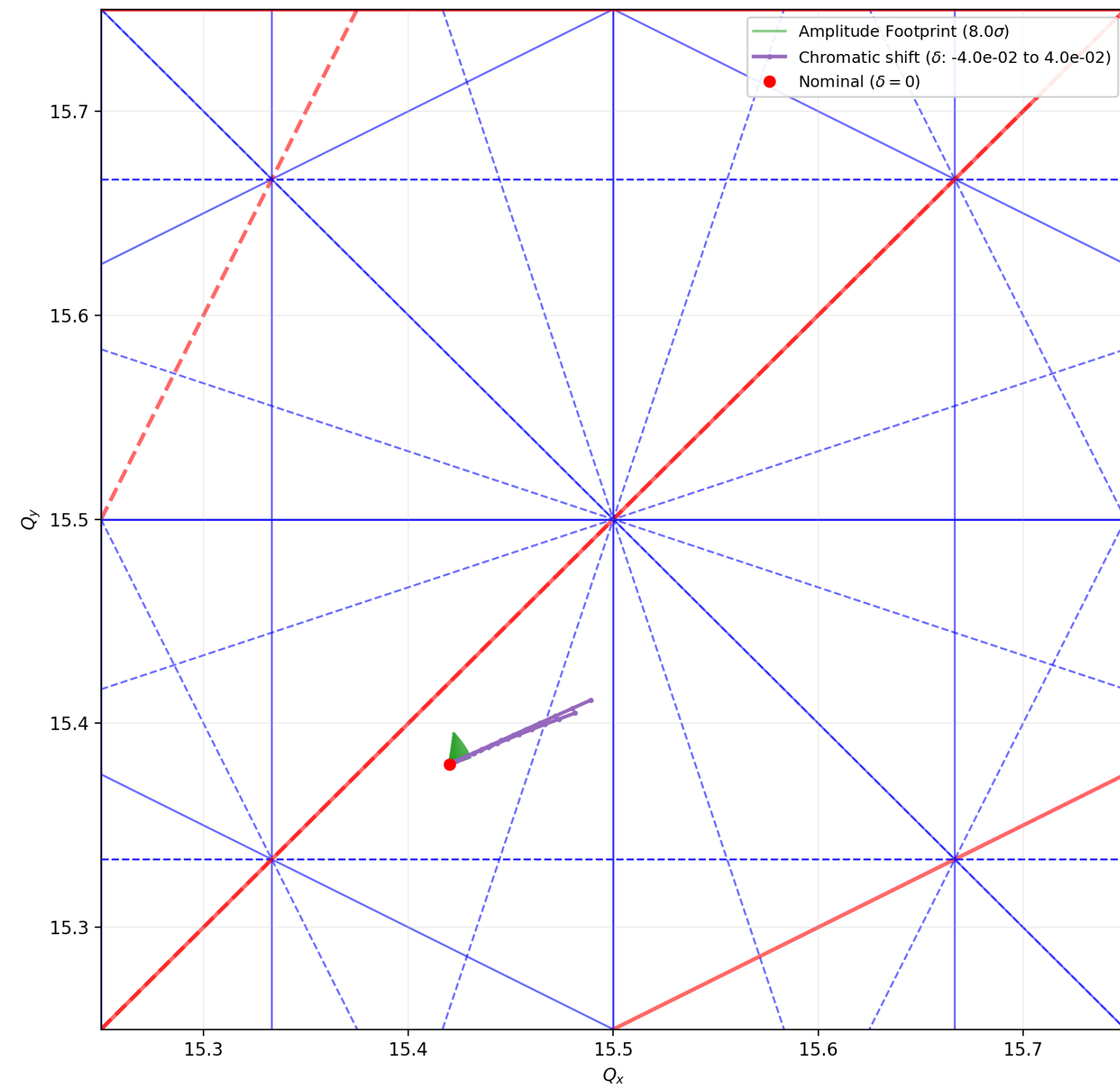
- **Ring circumference:** 215.4m
- **Equilibrium emittance:** 30.4nm
- **Polarization build-up time:** 12.41 minutes
- **Maximum bending field:** 1.47T

Ring description

- Three-fold periodicity
- 8 focusing, and 8 defocusing, sextupoles per half-arc (2 families)
- Each arc has 8 FODO cells followed by a dispersion suppressor and a triplet
- All dipoles are the same length, and an extra quadrupole has been added in the dispersion suppressor



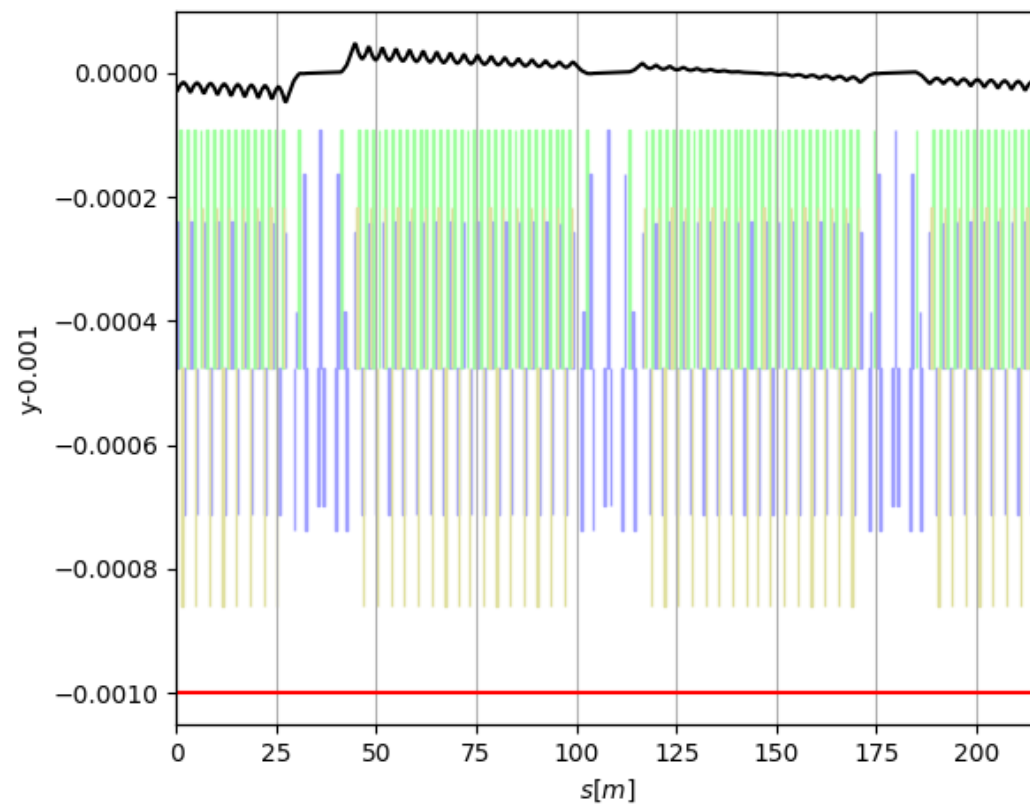
Working point



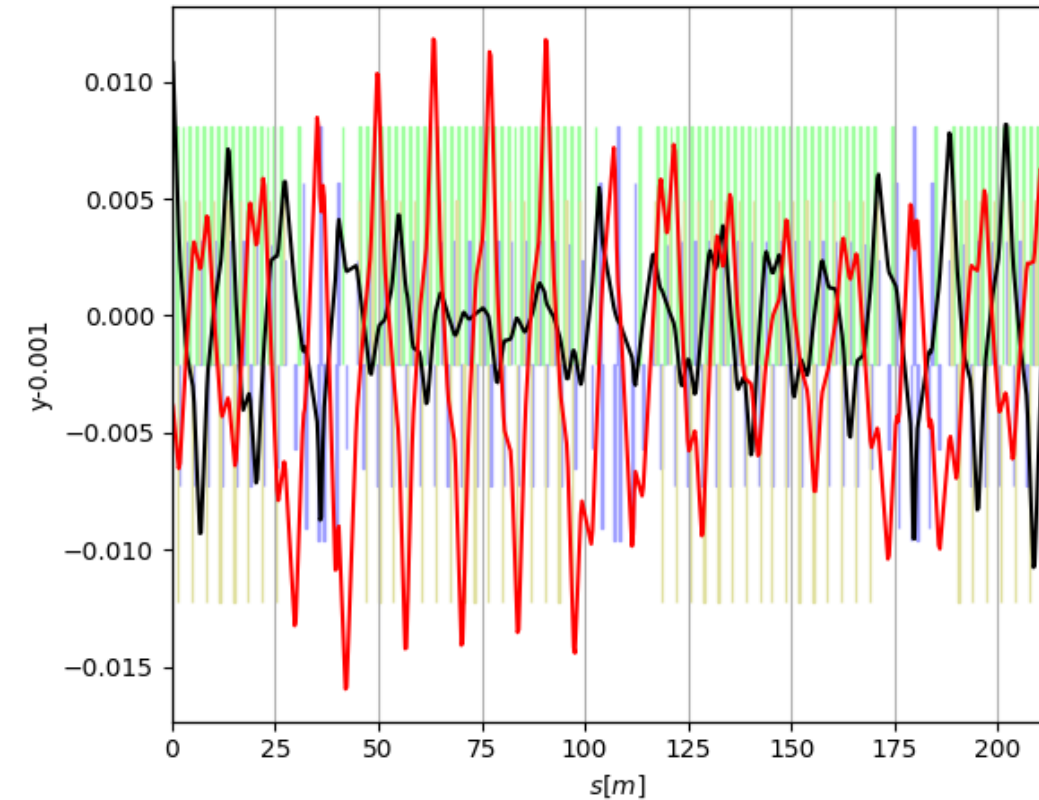
- **Selected working point: 15.42, 15.38**
- Doesn't cross any systematic resonances up to 4th order
- Work to be done to reduce nonlinear terms

Misalignments and orbit correction

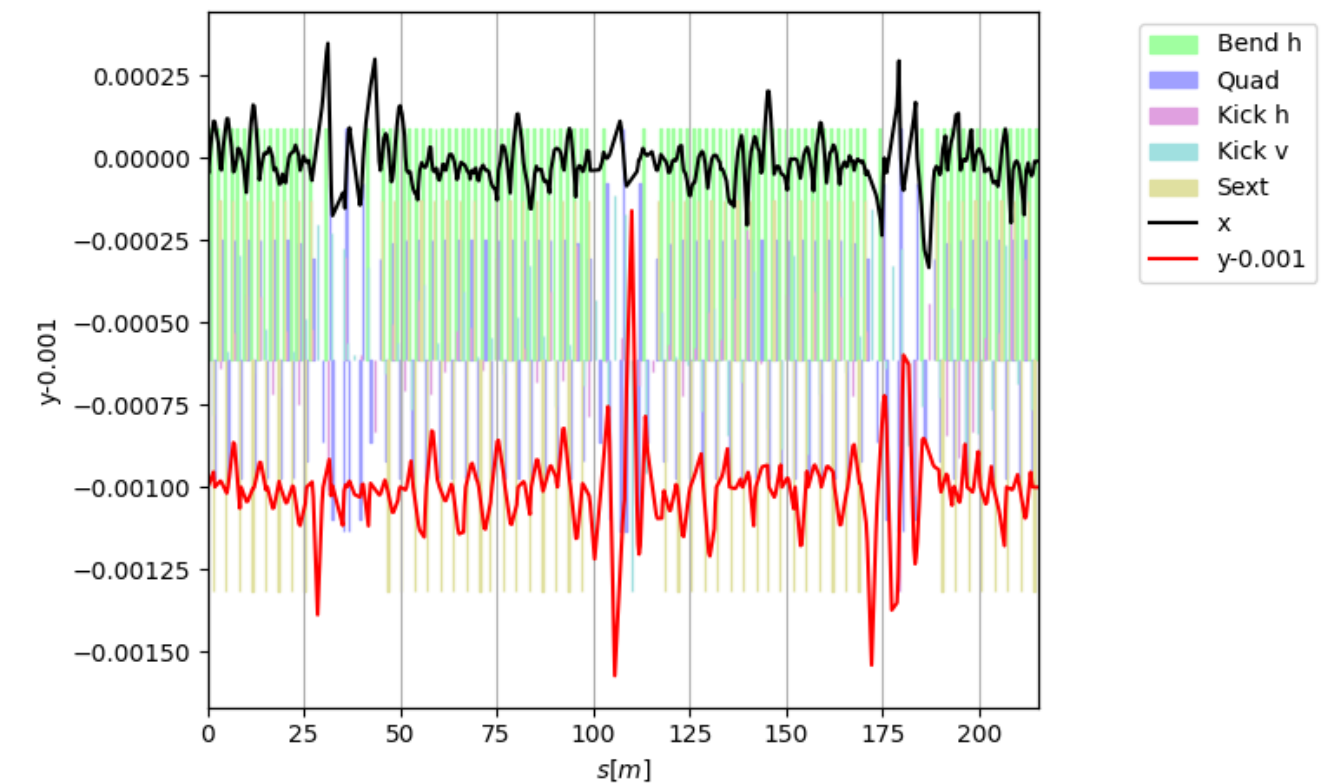
Perfect



Misaligned



Corrected



Misalignments

Applied to all bends, quadrupoles, sextupoles, and correctors:

- **Translational misalignments** (horizontal, vertical, longitudinal): 0.25mm
- **Rotational misalignments** (roll, pitch, yaw): 0.25mrad
- **Relative field error**: $\pm 0.1\%$ relative error applied to each magnet's integrated normal and skew strengths

Currently unwanted multipolar components and BPM errors have not been modelled

Corrections

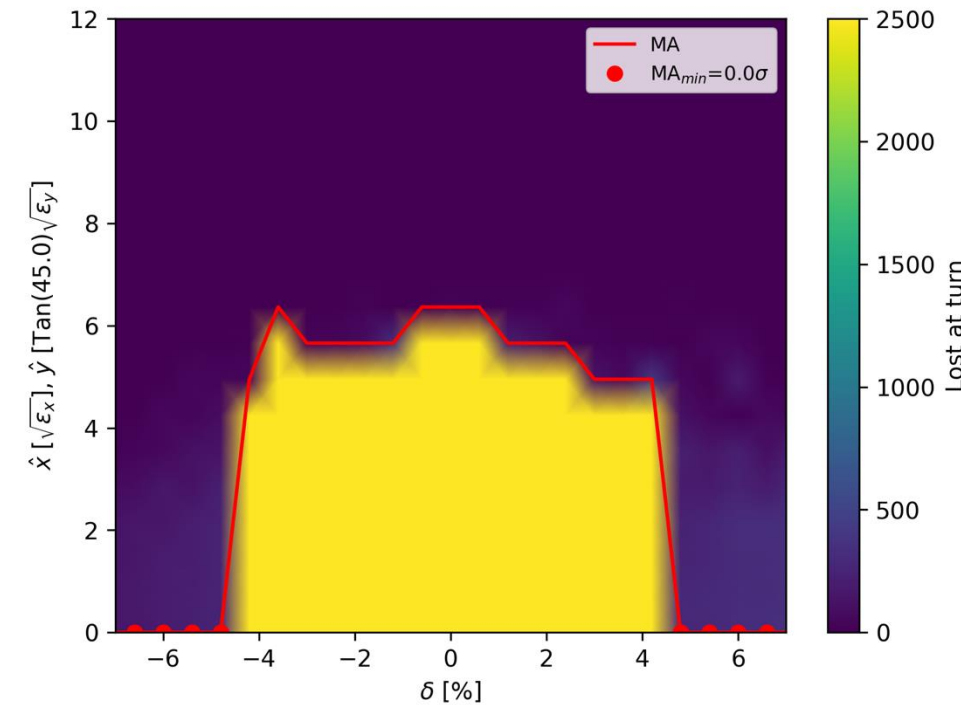
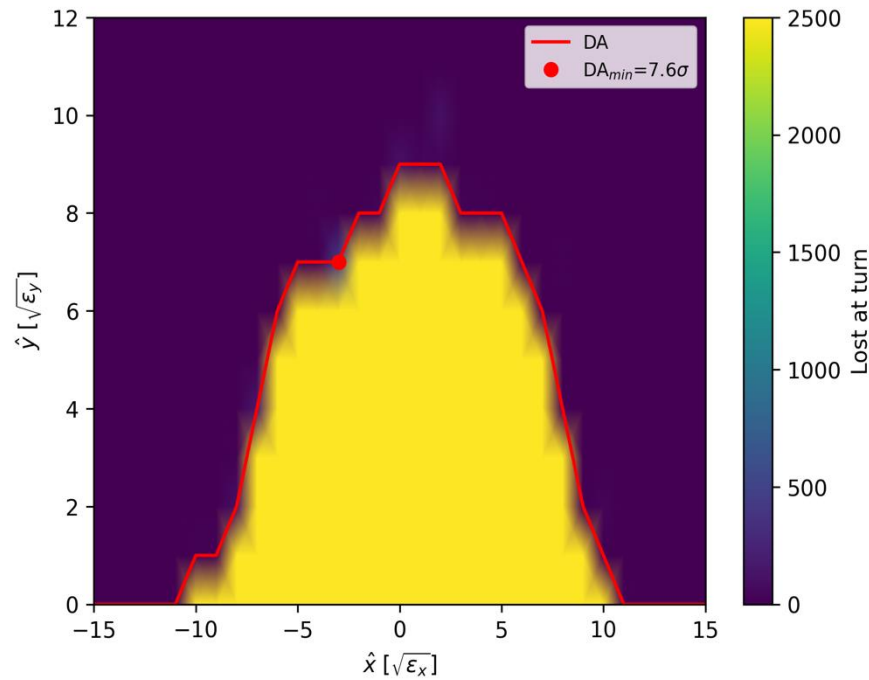
- Horizontal correctors placed around every focusing quadrupole, and vertical correctors placed around every defocusing quadrupole
- BPMs placed around every quadrupole
- Using SVD orbit correction scheme with 15/30 horizontal eigenvectors and 15/30 vertical eigenvectors

Dynamic aperture and momentum acceptance

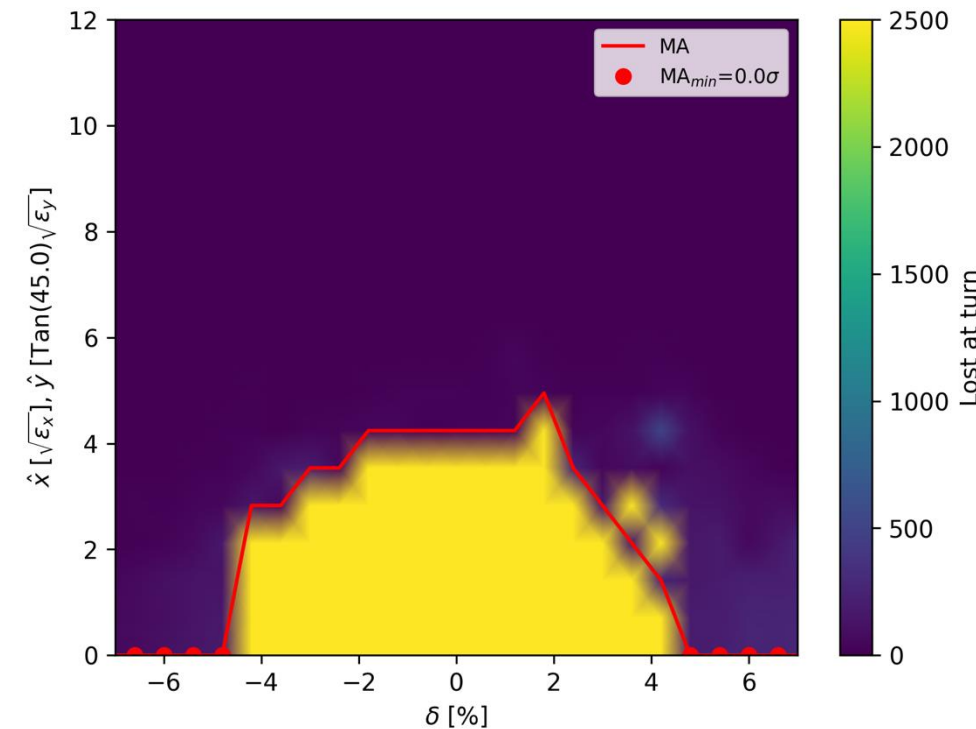
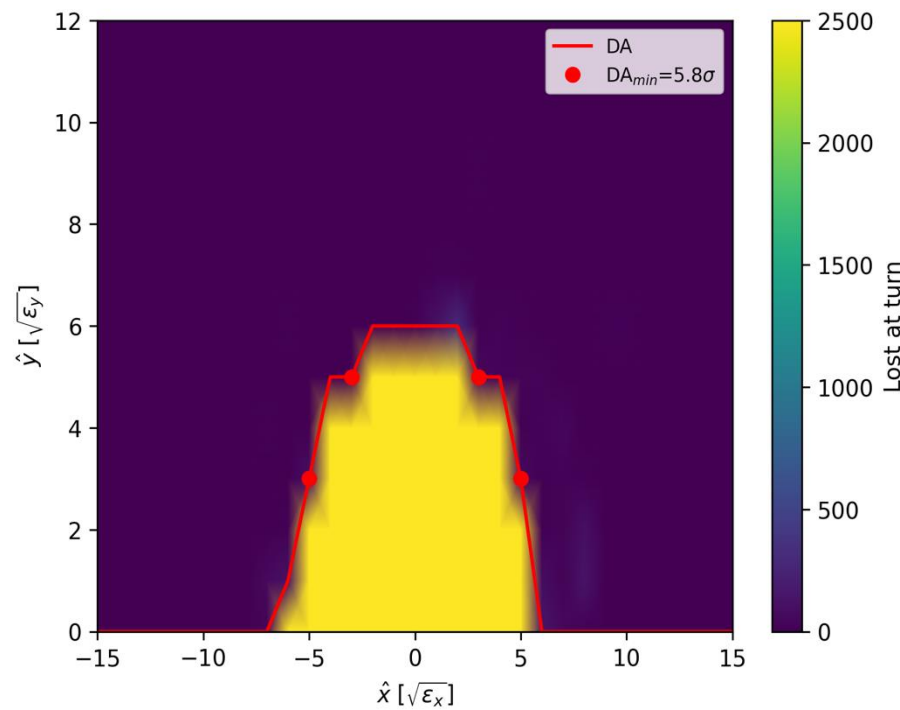
Dynamic aperture

Momentum acceptance

Perfect lattice

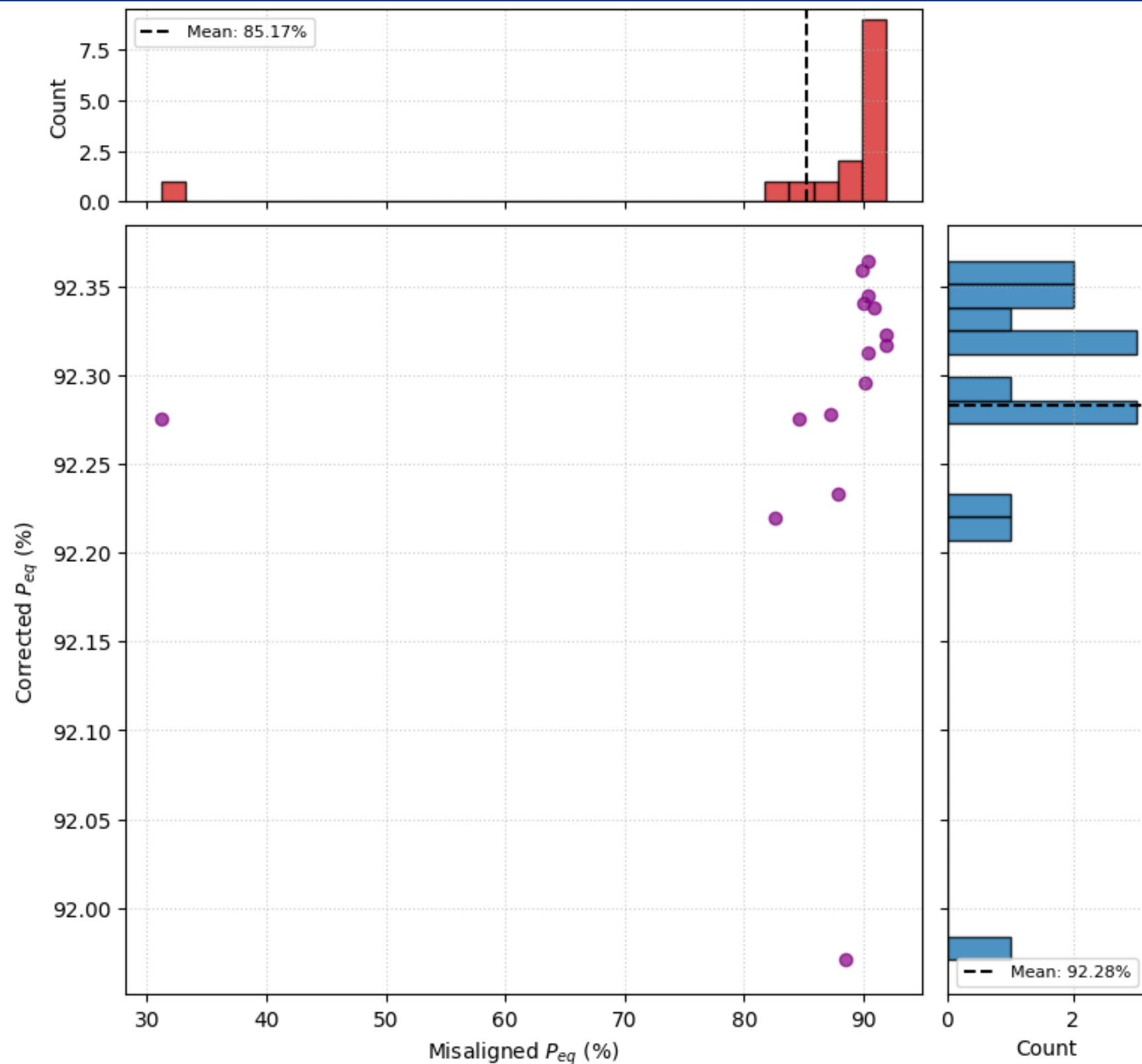


Corrected lattice



- DA/MA studies were run for 2500 turns, tracking 1000 particles
- The emittances used were calculated from the positron linac macroparticle distribution:
 - Horizontal emittance = 1.35 μm
 - Vertical emittance = 1.42 μm
- The corrected lattice was run for a single misalignment seed (in future it will be done for multiple cases)
- $DA_{\min}$ reduces from 7.6σ to 5.8σ
- MA remains high at approx. $\pm 5\%$
 - The steep edges suggest its limited by the RF bucket
 - The current RF voltage is 16 MeV

Spin tracking and equilibrium polarization

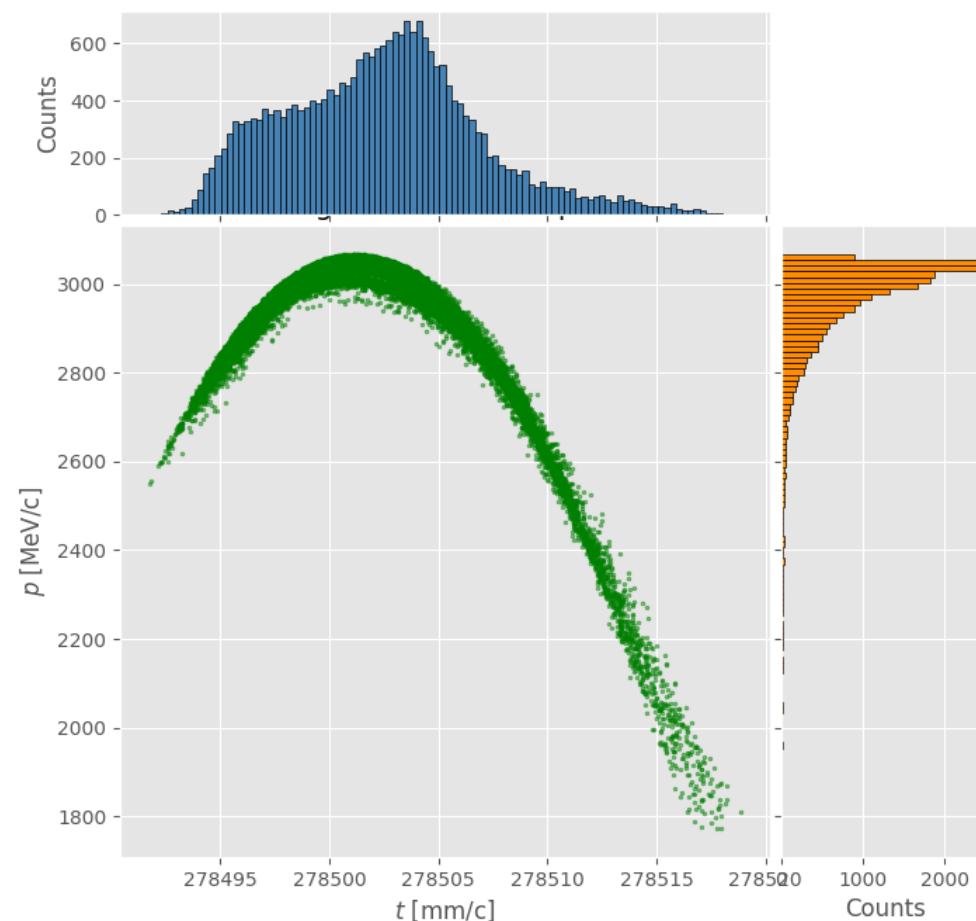


- Spin was tracked for 20 misalignment seeds, and their corresponding corrections
- Even badly performing misalignment seeds showed a significant improvement after correction (e.g. polarization increased from 31.25% to 92.27%)
- The average equilibrium polarization for corrected cases is 92.28% (approaching the asymptotic limit of 92.38%)

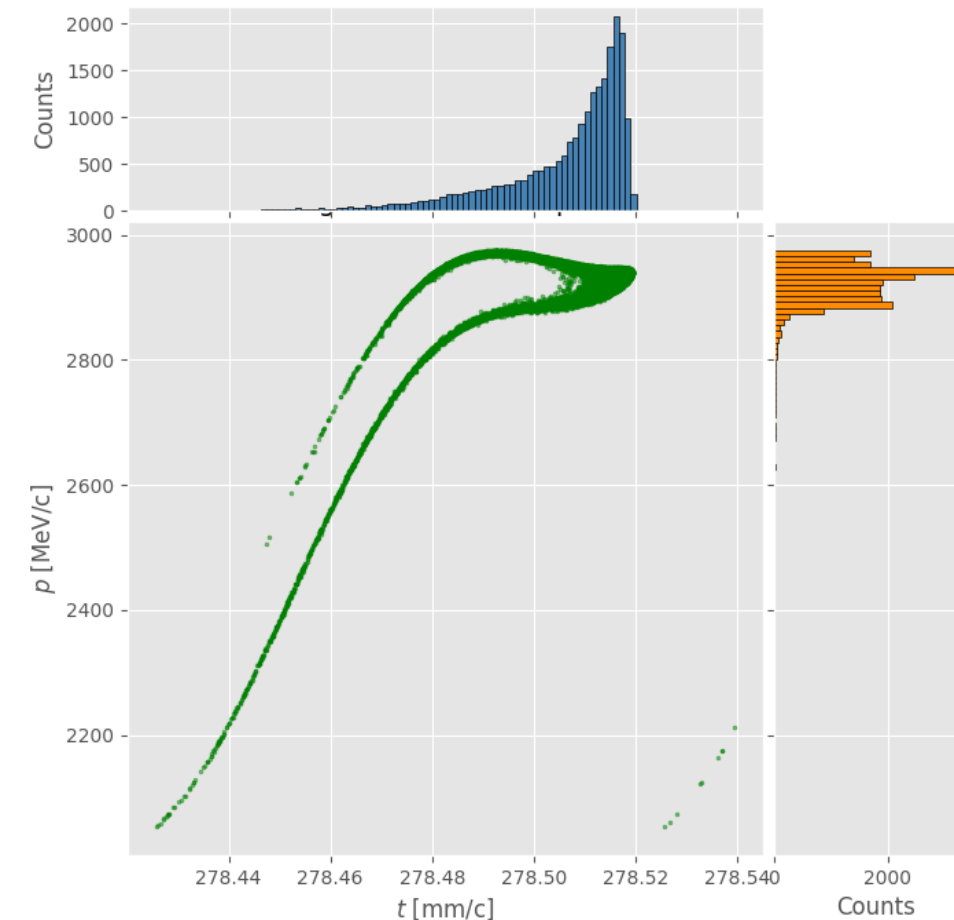
NOTE: The y-axis scale is much smaller than the x-axis in order to show the histograms

Injection efficiency

Original longitudinal distribution

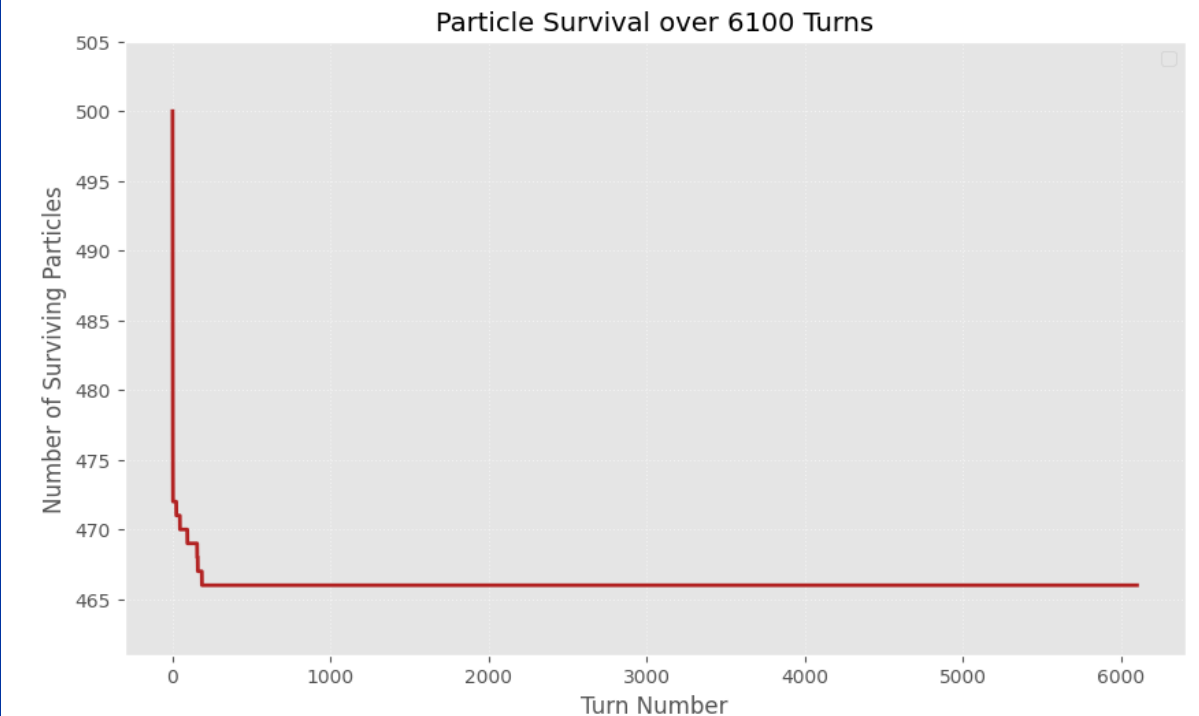


Compressed longitudinal distribution

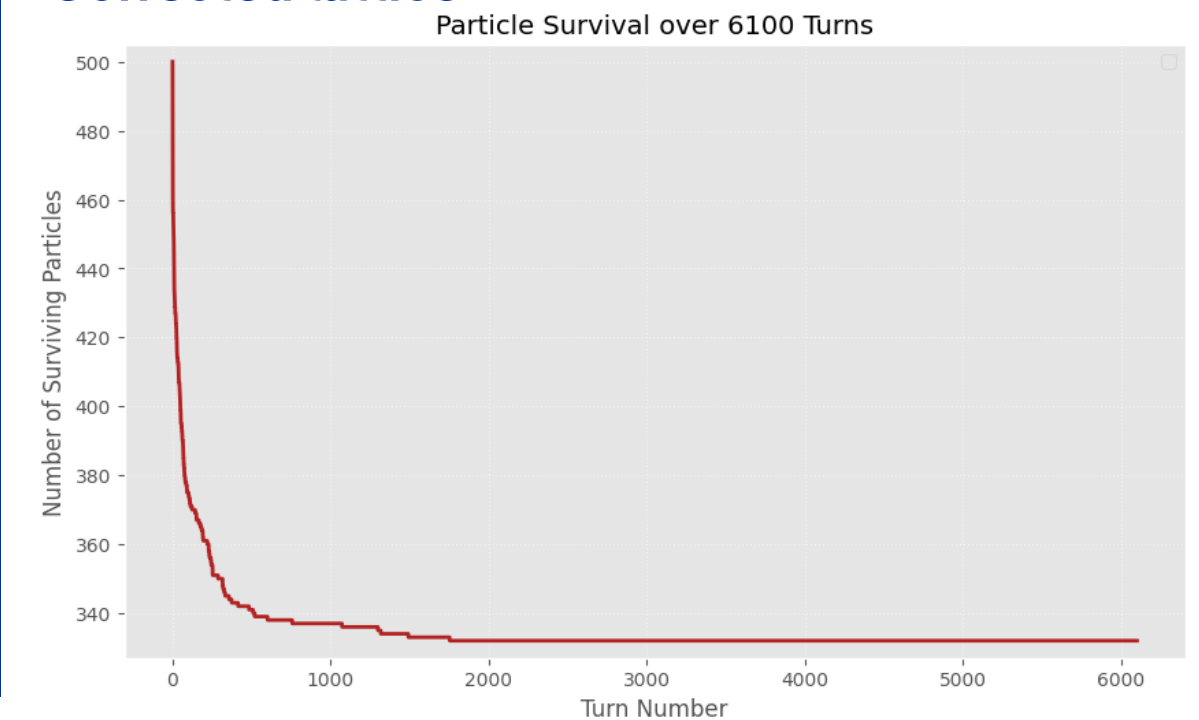


- The positron distribution from the linac was put through an energy compressor simulation to reduce energy spread
- The tails were sheared, retaining **96.7%** of the original particles
- An energy scan was performed to determine the optimal injection energy, in this case **2.95GeV**
- A random subset of 500 particles was tracked over 6100 turns
- The perfect lattice had an injection efficiency of **93.2%**, which reduced to **66.4%** after misalignments and correction

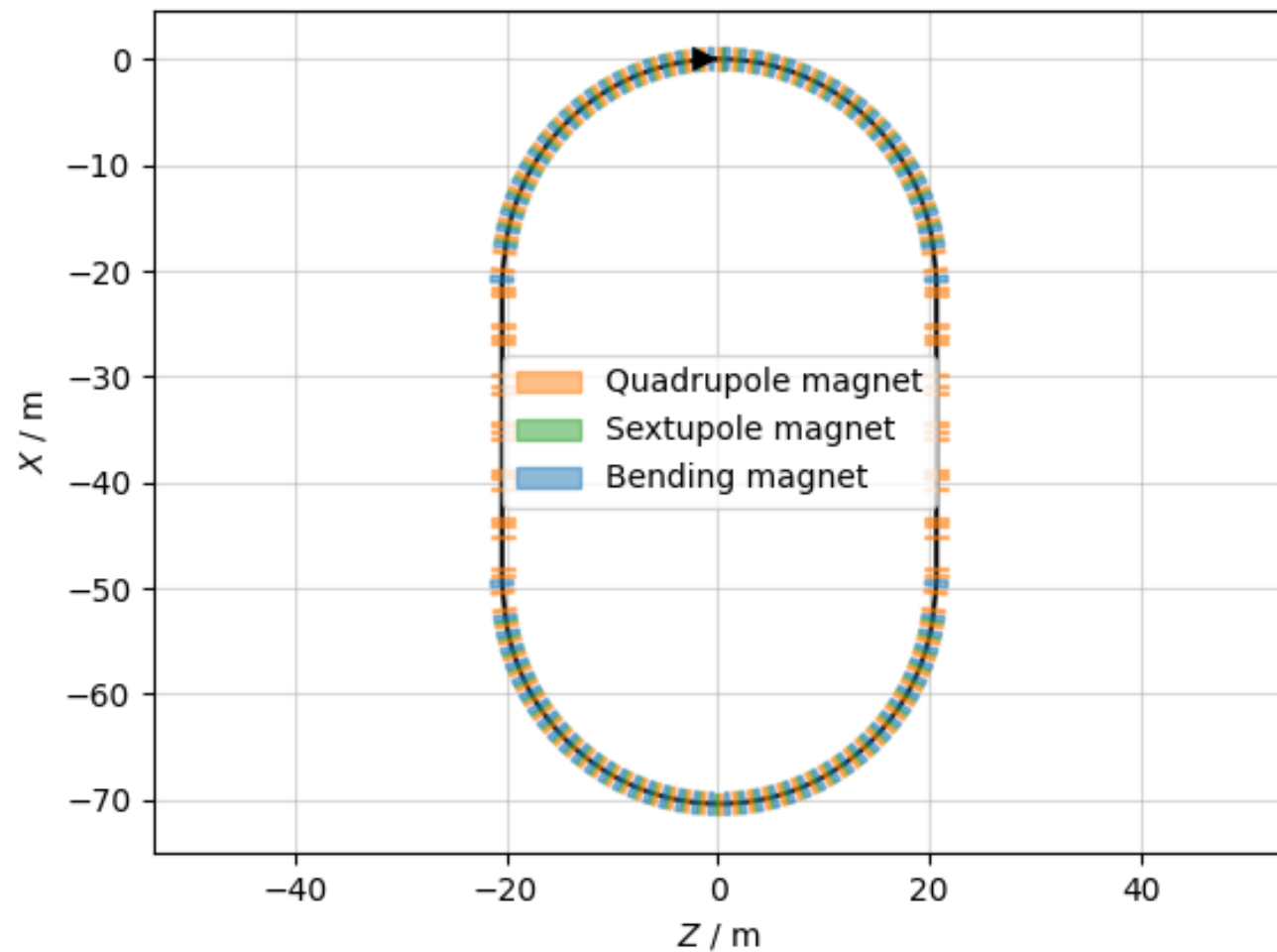
Perfect lattice



Corrected lattice



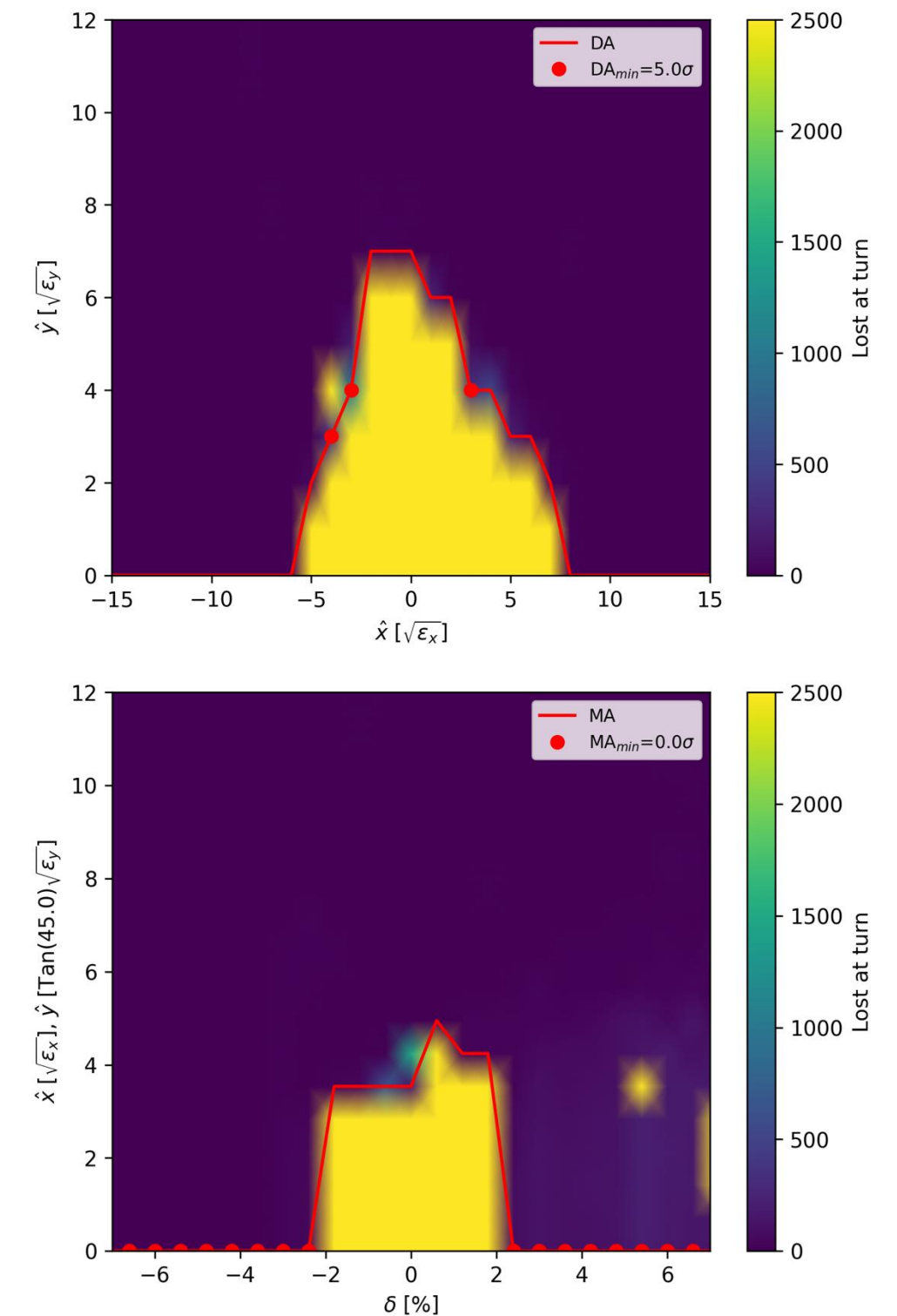
Racetrack design



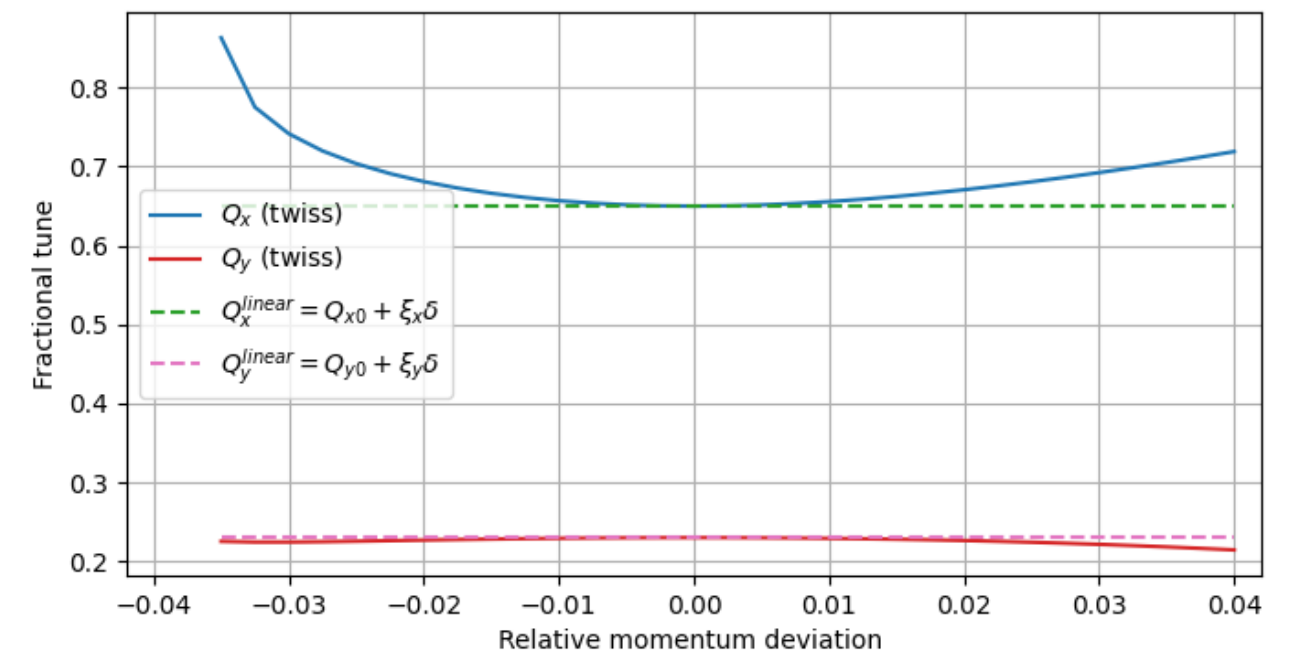
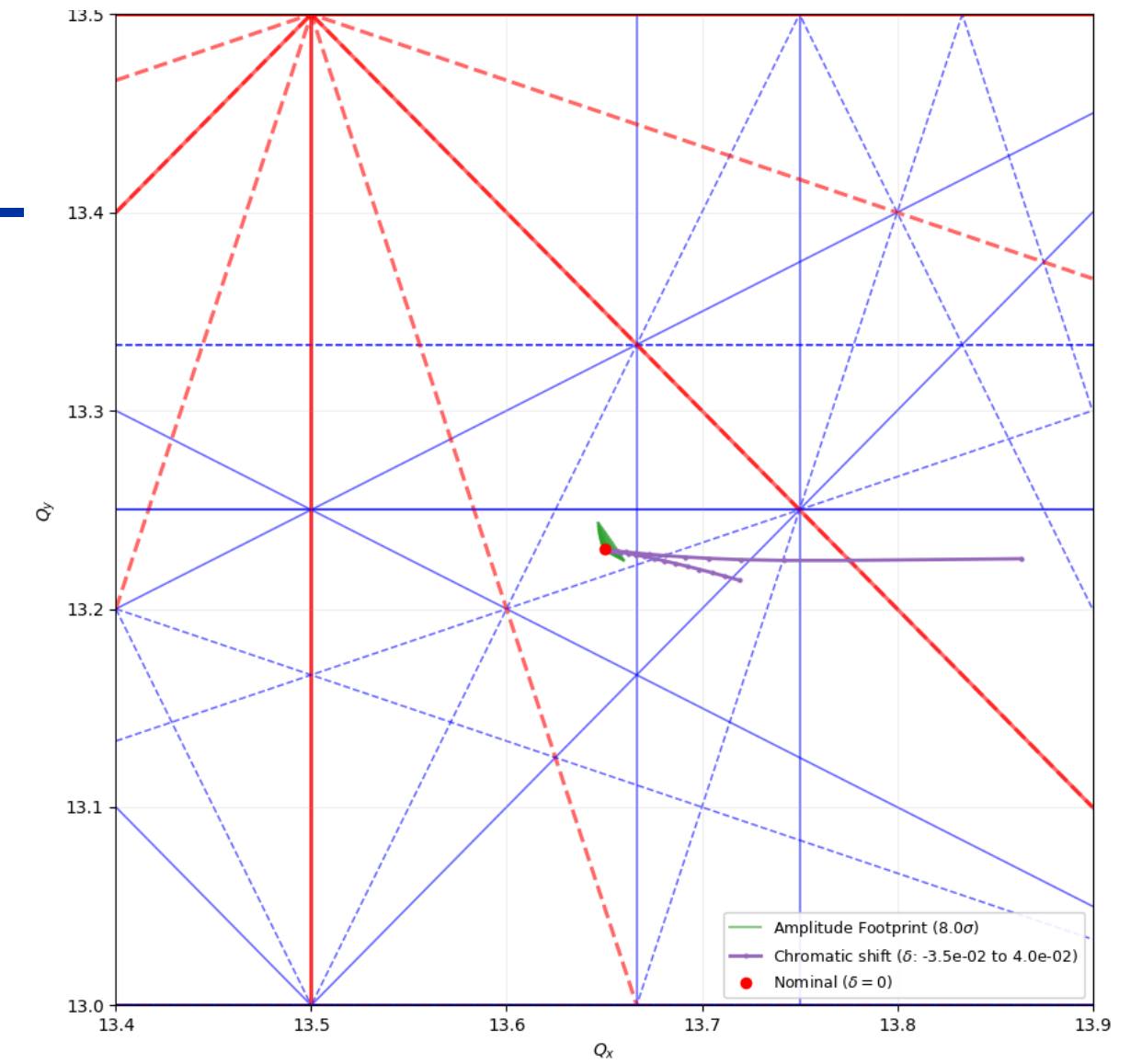
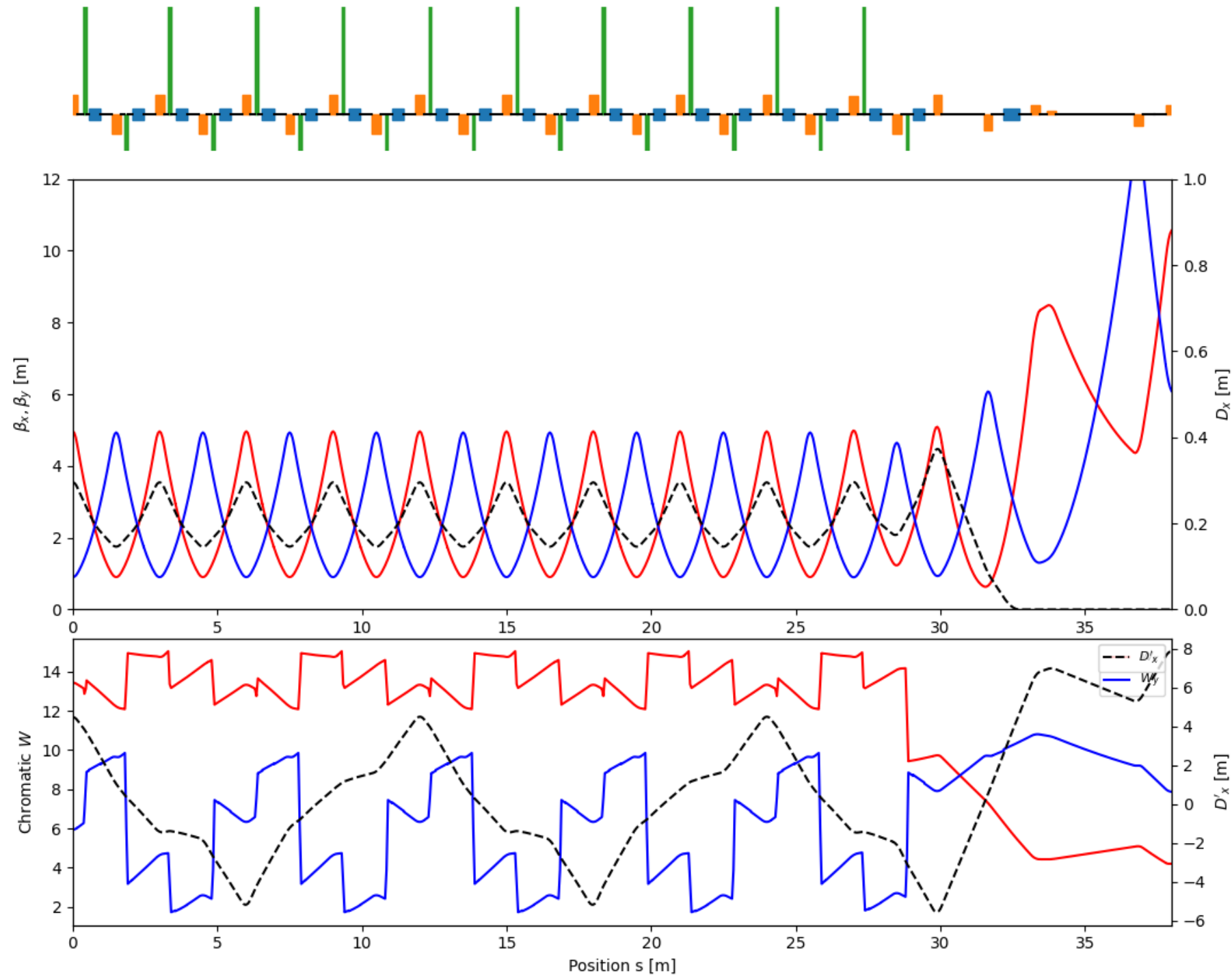
Ring parameters

- **Ring circumference:** 187.4m
- **Equilibrium emittance:** 47.3nm
- **Polarization build-up time:** 7.59 minutes
- **Maximum bending field:** 1.75T

- Each arc has 10 FODO cells, a dispersion suppressor, and 3 triplets
- Currently, there is a focusing sextupole around every focusing quadrupole, and a defocusing sextupole around every defocusing quadrupole in the FOCO arc cells
- DA/MA aren't as large as baseline, but still promising
- Optimisation of the sextupole layout and working point placement need to be explored

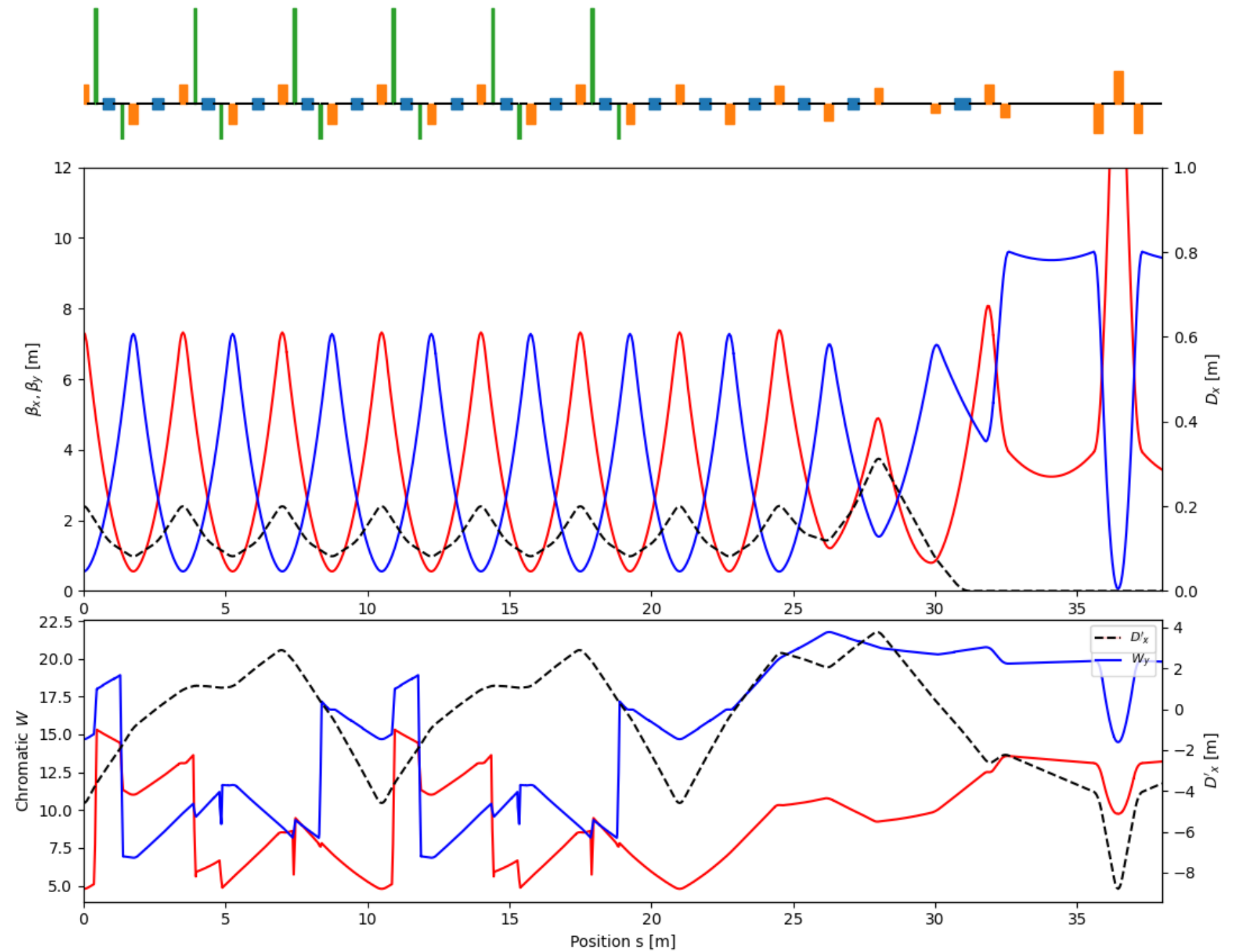
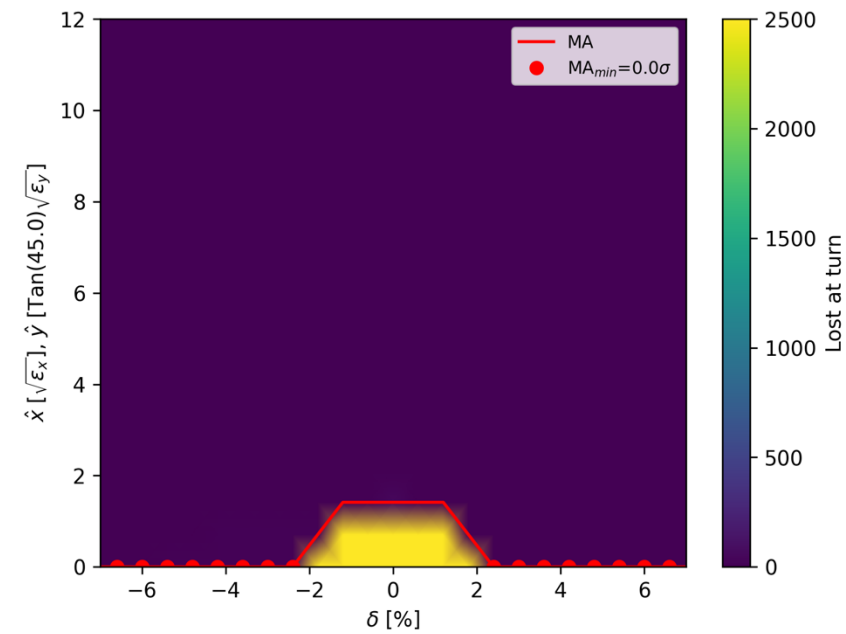
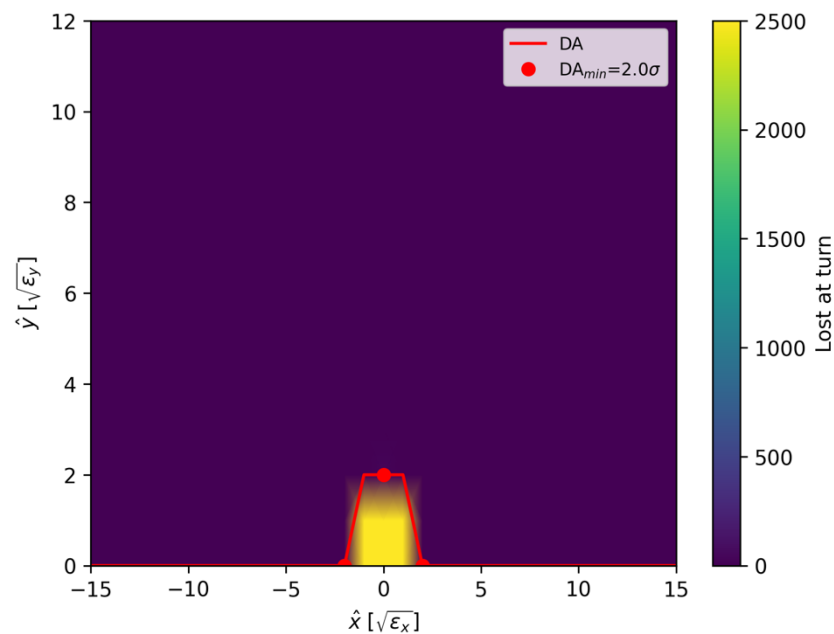


Racetrack design



120° phase advance

- Uses the same base ring but with increased phase advance
- This is being explored to reduce equilibrium emittance
- Currently 6 focusing, and 6 defocusing sextupoles, in each half-arc
- There is an extremely small vertical waist in the triplet that needs to be cured
- DA/MA are very small
- Once again, optimisation of the sextupole layout and working point placement need to be explored



Conclusion and next steps

Conclusion

- Baseline ring available that produces large momentum acceptance and sufficient equilibrium polarization for multiple seed cases
- Alternative designs are being explored but need to be optimised to see if they are viable options

Next steps

- How tightly packed ring elements can be integrated into the injector complex
- Explore options for longer rings:
 1. Keep the energy the same and add asymmetric wigglers into the straights
 2. Increase the bending radius and accelerate the beam